

Graph Neural Networks for Electricity Load Forecasting: A Comparative Study of GCN and GAT Architectures on Florida County Data

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Abstract—Accurate, short-term electricity load forecasting is critical for grid stability, resource scheduling, and market operations. Most forecasting models treat each location independently, so they miss the useful fact that nearby regions tend to have similar demand patterns. In this paper, we build a graph of Florida’s 67 counties where edges connect geographically close counties, and we use that graph to improve hourly load forecasting. We compare three models: a standard Feed-Forward Neural Network (FF-NN) that ignores geography, a Graph Convolutional Network (GCN), and a Graph Attention Network (GAT). All models are trained on 2021–2022 load data and evaluated on 2023 data using MAPE and RMSE. Our results show that GCN graph models outperformed the baseline feed-forward model by about 33%. We published our code and framework and made it available at: <https://github.com/pa233796/FLpowerGNN>

Index Terms—Graph Neural Networks, GCN, GAT, Electricity Load Forecasting, Spatial Modeling, Florida Power Grid.

I. INTRODUCTION

A. Background and Motivation

Electricity is unique among commodities in that it must be produced and consumed at almost the same instant – large-scale storage is still not practical. This means power companies must predict how much electricity will be needed hours in advance to plan generation accordingly.

The challenge has grown over the past decade. The spread of electric vehicles and rooftop solar is reshaping demand in ways older models did not anticipate. Climate change is producing hotter summers and more frequent extreme weather events, which can push demand to unexpected highs in a short window of time.

One opportunity that is often overlooked is the spatial correlation between neighboring regions. On a hot summer afternoon in one Florida county, the county next door is almost certainly just as hot and drawing just as much power for air conditioning. If a model can pass that kind of information between neighbors, it should make better predictions than one that treats each county in isolation.

B. Problem Statement

This project asks a direct practical question: does modeling Florida counties as a spatially connected graph, rather than as 67 independent time series, lead to better hour-ahead electricity load forecasts? And between the two graph models we test, does the GAT’s learnable attention offer a meaningful advantage over the GCN’s simpler averaging?

C. Our Contributions

- We adapt the GNN forecasting framework of Campagne *et al.* [1] to a U.S. county-level setting for the first time.
- We construct a geographic graph of Florida’s 67 counties using a Gaussian distance kernel and apply it to real hourly load data from the NREL OEDI dataset [2].
- We run a clean three-way comparison of FF-NN, GCN, and GAT under identical hyperparameters and the same 2021–2022 training / 2023 test split.

II. RELATED WORK

A. Classical Forecasting Methods

For many years, Generalized Additive Models (GAMs) were the go-to tool for electricity load forecasting [3]. They decompose demand into smooth components for time of day, day of week, and temperature, then add them up. They are interpretable and reliable under normal conditions, but they require manual feature design and treat each node independently. With evolving and growing decentralized power distribution networks, literature in the state-of-the-art suggests that a method capable of modeling decentralized electricity networks, such as graph neural networks, may be better suited for this task [4].

B. Deep Learning for Load Forecasting

Recurrent Neural Networks, and particularly Long Short-Term Memory (LSTM) models, have shown strong results by capturing temporal correlations in load series [5]. However,

these architectures process each node independently and do not model inter-node interactions, limiting their ability to exploit spatial structure.

C. Graph Neural Networks for Energy System

GNNs were first developed for social networks and chemistry, but they transfer naturally to any domain with a relational structure. Traffic forecasting, where cars move along a road network, was an early success story and is structurally similar to electricity forecasting along a grid [6].

Campagne *et al.* [1] published one of the most comprehensive benchmarks of GNNs for electricity load forecasting. They tested GCN, GraphSAGE, APPNP, and GAT on French regional data and UK residential smart-meter data. Their findings shown that graph models consistently outperformed conventional baselines. Our work follows their methodology closely and applies it to the counties of the U.S. state of Florida.

III. METHODOLOGY

A. Building the Graph

We represent each of the 67 Florida counties as a node. Two counties get an edge between them if they are geographically close. We measure closeness using a Gaussian kernel applied to geodesic distance:

$$d(u, v) = \exp\left(-\frac{\text{dist}(u, v)^2}{\sigma_b^2}\right), \quad (1)$$

where $\text{dist}(u, v)$ is the straight-line distance between county centroids and σ_b is set to the median of all pairwise distances (approximately 164.18 miles for Florida). This gives a weight close to 1 for nearby counties and close to 0 for distant ones.

To keep the graph sparse, we drop any edge whose weight falls below a threshold $\lambda = 0.2$:

$$W_\lambda = d(u, v) \cdot \mathbf{1}[d(u, v) \geq \lambda]. \quad (2)$$

The result is a connected but sparse graph where each county is linked to a handful of nearby neighbors, following the approach in previous studies.

B. Features and Targets

Each node's input at time t is its own load from the previous 24 hours, stored as a vector $\mathbf{x}_{v,t} \in \mathbb{R}^{24}$. The target $y_{v,t}$ is the load at the next hour. This is a sequence-to-one setup where the model uses the previous day's pattern to predict the immediate next step. All load values are normalized to $[0, 1]$ with `scikit-learn`'s `MinMaxScaler` utility function.

C. Train and Test Split

We train on reported hourly demand profiles recorded from January 2021 through December 2022 and test against data from 2023. The two windows do not overlap, so evaluation reflects genuine out-of-sample generalization.

D. Evaluation Metrics

We report two metrics, matching the evaluation in Campagne *et al.* [1]. Mean Absolute Percentage Error (MAPE) expresses error as a percentage of the true value, making it easy to interpret:

$$\text{MAPE} = \frac{1}{T} \sum_{t=1}^T \left| \frac{\hat{y}_t - y_t}{y_t} \right| \times 100\%. \quad (3)$$

Root Mean Square Error (RMSE) is in physical units (megawatts) and penalizes large errors more than small ones:

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t)^2}. \quad (4)$$

Both metrics are computed on the aggregated statewide load, obtained by summing predictions across all 67 counties. This combination of metrics is commonly used in prior works and makes for an adequate evaluation of forecasting accuracy and stability.

IV. MODEL ARCHITECTURES

A. How GNNs Work

The core idea of a GNN is straightforward: every node starts with its own feature vector. At each layer, it collects information from its neighbors, combines it with its own current state, and updates itself. Stack a few layers and each node ends up informed by a wider neighborhood. The general update rule is:

$$\mathbf{h}_v^{(\ell+1)} = \text{UPDATE}\left(\mathbf{h}_v^{(\ell)}, \text{AGG}\left\{\mathbf{h}_u^{(\ell)} : u \in \mathcal{N}(v)\right\}\right), \quad (5)$$

where $\mathbf{h}_v^{(0)} = \mathbf{x}_v$. The difference between GCN and GAT lies entirely in how they compute the AGG step.

B. Baseline: Feed-Forward Neural Network (FF-NN)

The FF-NN takes each county's 24-hour history and passes it through two fully connected layers, each consisting of 32 hidden units and a ReLU activation function between the first and second layers before producing a single load prediction. It sees no graph information at all. If the graph models cannot surpass this, adding graph structure is not worth the effort.

C. Graph Convolutional Network (GCN)

The GCN takes the graph and features described in section III-A and III-B and passes it through two convolution layers with 32 hidden channels and ReLU activation each. A dropout layer with $p = 0.2$ is placed between the convolution layers. The output of the second convolution layer is flattened and passed to a single linear fully-connected layer for the final regression task. The network uses the graph convolutional operator from [7].

D. Graph Attention Network (GAT)

The GAT is otherwise identical to the GCN in architecture except that it uses the graph attentional operator from [8].

V. IMPLEMENTATION

A. Dataset

Power load data comes from the NREL OEDI hourly county-level demand dataset [2], which covers the entire contiguous United States from 2016 through 2023. We extract Florida data from 2021 to 2023. County geographic distances for the graph are provided by the 2010 census [9].

B. Graph Construction Results

Before building the graph we verified that all 67 Florida counties were present in both the load time-series and the distance datasets, confirming 4,422 directed county pairs with no missing entries. Applying the Gaussian kernel (Eq. 1) with $\sigma = 164.18$ miles and then thresholding at $\lambda = 0.2$ (Eq. 2) reduces the edge count from 4,422 to 1,391, removing roughly 31% of the weakest long-distance connections.

Figure 1 shows the raw fully connected network. Every county pair has an edge, creating a dense mesh that would dilute local demand signals during GNN message passing.

Figure 2 show the graph after applying $\lambda = 0.2$. The uniform-edge view (Fig. 2) highlights which county pairs remain connected after pruning.

1. Before Thresholding (Fully Connected)

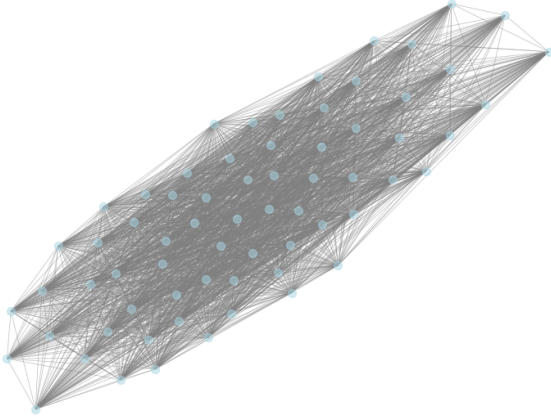


Fig. 1. Florida county graph **before** thresholding. 2211 edges in total.

C. Software Stack

All code is written in Python, organized into Jupyter notebooks, and runs on workstations with GPU acceleration. We use PyTorch for model training and PyTorch Geometric [10] for the graph convolution layers. pandas and numpy handle data loading and preprocessing, and matplotlib produces the plots, and certain utility functions from scikit-learn are also used. The code notebooks and required files to run the workflow are available on GitHub: <https://github.com/pa233796/FLpowerGNN>

2. After Thresholding ($\lambda=0.2$) - Uniform Edges

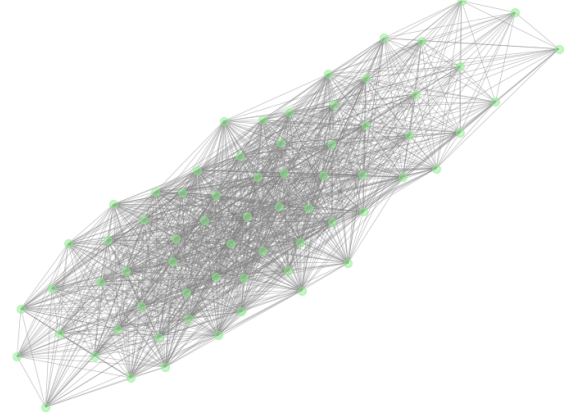


Fig. 2. Florida county graph **after** thresholding ($\lambda = 0.2$) with uniform edge thickness. 1391 edges in total.

Model	batch_size	n_layers	hidden_channels	lr
FF	32	2	32	0.0001
GCN	32	3	32	0.01
GAT	32	3	32	0.01

TABLE I
MODEL HYPERPARAMETERS

D. Training Details

The hyper-parameters used for the different models are summarized in Table I. Parameters are optimized using the Adaptive Moment Estimation (Adam) optimizer minimizing the Mean-Square-Error (MSE) loss function. Models are trained using a standard mini-batch stochastic gradient decent method.

VI. RESULTS

A. Training and Validation Loss

Training and validation loss across epochs are summarized in Figures 3 to 5. All models were trained for 500 epochs.

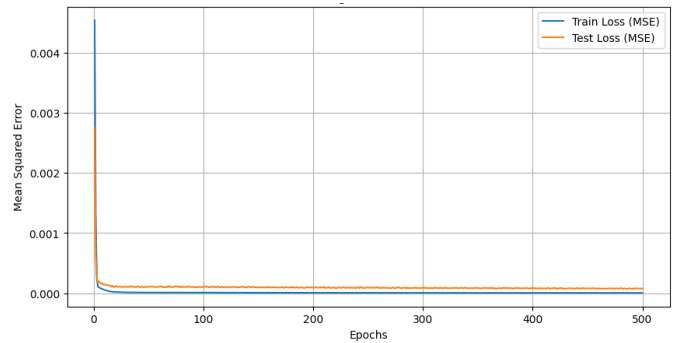


Fig. 3. Train and Test Loss for the FF network model

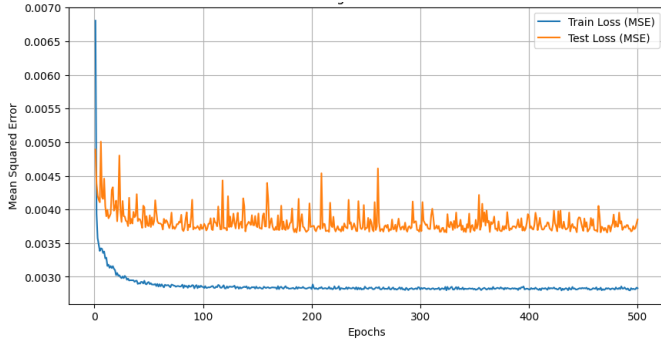


Fig. 4. Train and Test Loss for the GCN model

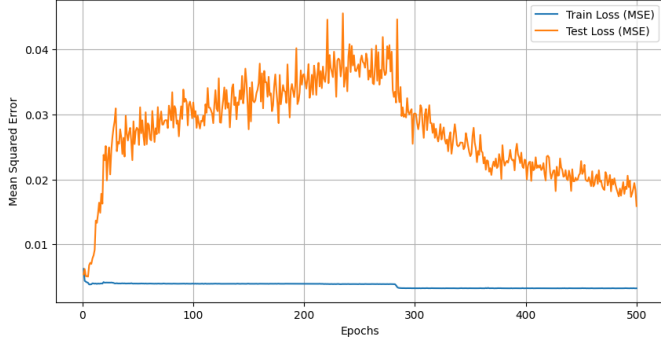


Fig. 5. Train and Test Loss for the GAT model

B. Forecast Accuracy

We report the MAPE and RMSE of the models in Table II. These metrics give insights on how different model architectures influence accuracy in forecasting tasks. First, the baseline feed forward network obtained an adequate MAPE of 1.94% and a RMSE of 665.68 MW. The metrics of the baseline FF-NN model help us determine whether the additional effort of implement more complex graph-based strategies is worthwhile.

The GCN model achieved the best MAPE and RMSE (1.94% and 482.82 MW, respectively). This suggests that the modeling of spatial structures of the electrical network as graph embeddings results in an improvement of about 33% in accuracy for prediction and regression tasks in electricity demand forecasting.

Finally, the GAT model achieved the worst results compared to the baseline with a relatively high 5.09% MAPE and 1529.89 MW RMSE. This suggests that using complex graph models has diminishing returns and may negatively impact accuracy on simple training workflows without specialized hyperparameter tuning strategies.

C. Visual Forecast Comparison

In addition to training loss plots in Figures 3 to 4, a comparison between the predicted and ground-truth values for the first one hundred hours of the testing set time series is shown in Figures 6 to Figure 8.

Model	MAPE (%)	RMSE (MW)
FF	1.94	665.68
GCN	1.27	482.82
GAT	5.09	1529.89

TABLE II
PERFORMANCE PER MODEL

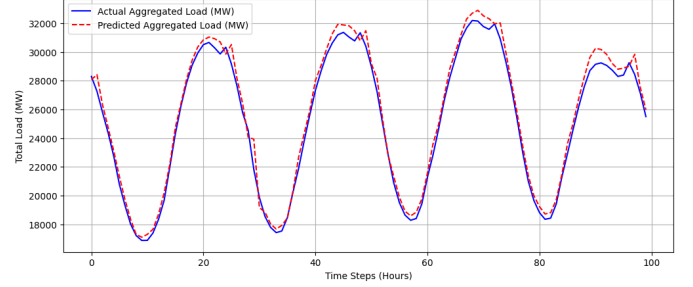


Fig. 6. Forecast vs Actual (First 100 test points)

D. Discussion

Our findings of graph models outperforming linear feed forward networks is consistent with the findings of Campagne *et al.* [1]. However, our results vary when we consider which graph model delivered the best results. While the previous paper found that GCN models perform slightly worse than the baseline and GAT perform slightly better, our results were the opposite: the GCN model outperformed both the baseline and GAT models, with GAT underperforming the baseline.

The causes for the discrepancy might vary from limited training or hyperparameter tuning setup to poor modeling of the GAT network. Visualizations from Figure 8 and the increasing validation loss during training for the GAT network shown in Figure 5 suggest that the GAT model tends to overfit. This result is coherent with that observation that “*in medium-scale settings with stable aggregated targets, architectural simplicity and strong inductive biases can match or even surpass the performance of more sophisticated designs.*” [1]

VII. CONCLUSION

We asked whether modeling Florida counties as a connected graph improves hour-ahead electricity load forecasting, and the answer was yes: the GCN model outperformed the non-spatial FF-NN baseline. And, while dissimilar to other studies,

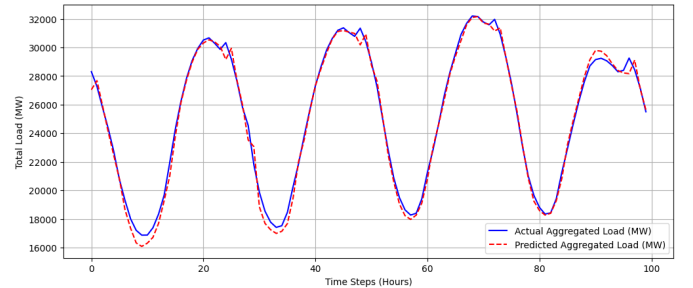


Fig. 7. Forecast vs Actual (First 100 test points)

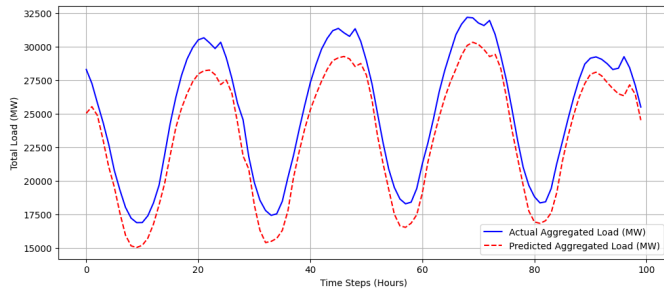


Fig. 8. Forecast vs Actual (First 100 test points)

we found that the GAT model underperformed, it showed promising results thanks to its attention mechanism that can adapt to changing spatial patterns. More focus is warranted on tuning hyper-parameters of GAT networks to leverage its complex but potentially better interpretability in electricity demand forecasting scenarios.

Future directions we would like to explore include adding weather covariates such as temperature and humidity as node features, testing longer prediction horizons, and analyzing how the GAT attention weights change across seasons to extract interpretable insights about which county relationships drive Florida's electricity demand.

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